

# Modeling wind resources in climate change scenarios for Catalunya

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## Motivation and goals:

Global warming is one of the most important environmental challenges that our society has to face nowadays. Administrations and policymakers are reacting to adapt and mitigate climate change. In Catalunya, strategies to increase the share of renewable energies in electric power production are being implemented, deriving in a spectacular growth of the installed wind farms, among others. However, future changes in climate could affect the availability of the wind resource in particularly vulnerable regions, such as the Mediterranean area.

This work derives changes in annual mean wind speeds and derived wind energy density implications for Catalunya, by dynamically downscaling three different Greenhouse Gases (GHG) emission scenarios: B1, A1B and A2, from present time to 2050, using WRF-ARW as a regional climate model at 10-km horizontal resolution.

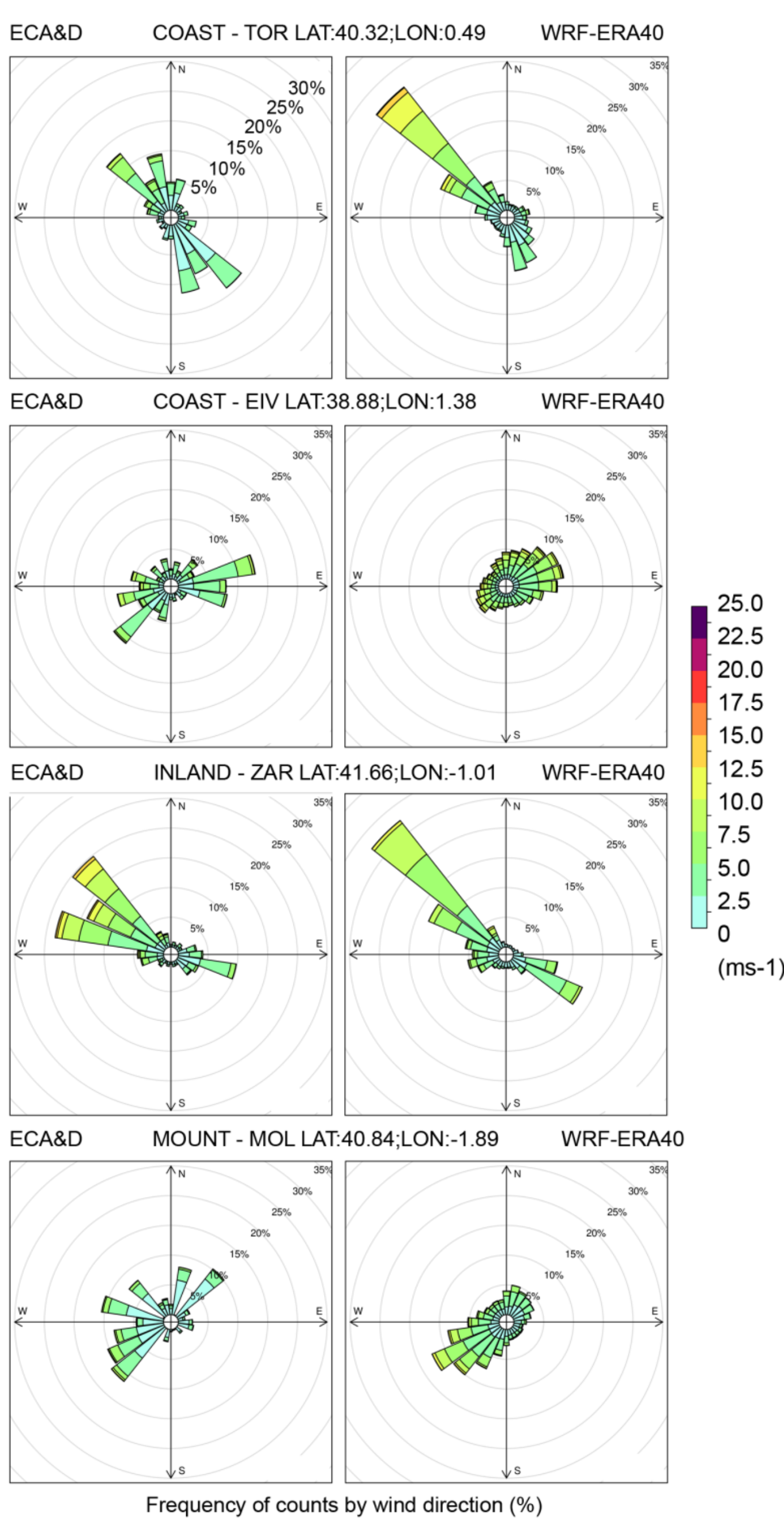
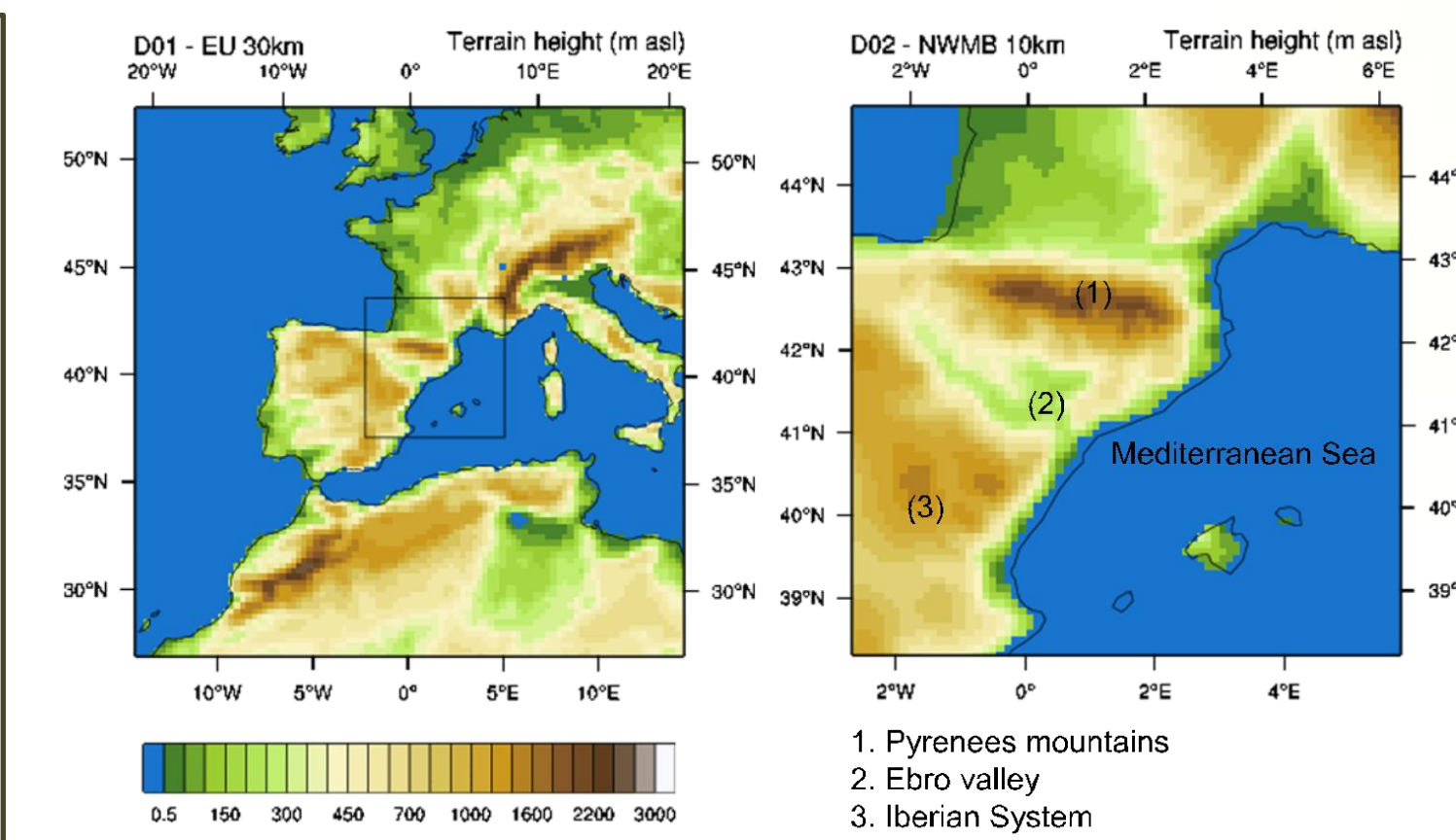
## Methods:

Two one-way nested domains are defined covering partially Europe (at 30-km horizontal resolution) and Catalunya (at 10-km), 33 vertical layers are used up to the top of the atmosphere.

WRF forced by the ERA40 (WRF-ERA40) reanalysis is run for the 1981-2000 period, in order to perform an assessment of the model capabilities to reproduce surface wind speed and direction in Catalunya.

Future projections for 2031-2050 are driven by the ECHAM5/MPI-OM atmosphere-ocean coupled GCM, considering three different GHG scenarios and two different initializations (run 1 and 3, WRF-EH1 and WRF-EH3).

Specific model configuration for the physics is selected according to previous studies and corresponds to the definition used in the ESCAT project, and hence in the 2nd Climate Change Report for Catalunya (SICCC).

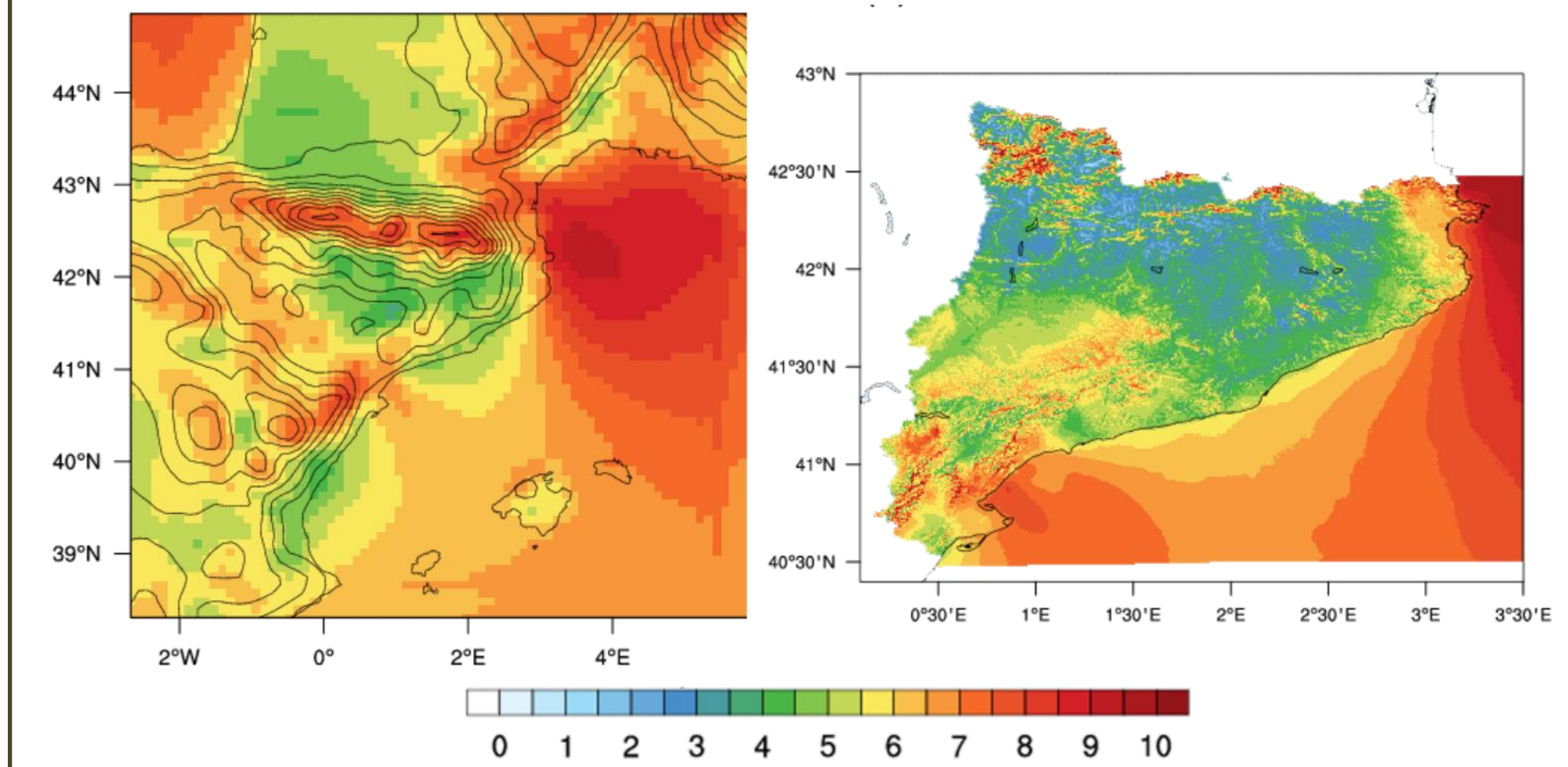


Wind roses as derived from daily wind speed and direction at 10 m agl from ECA&D stations data (left) and WRF-ERA40 (right) for 1981-2000 at four characteristic locations: coastal (two upper panels: Tortosa, TOR and Eivissa, EIV), inland (lower middle panels: Zaragoza, ZAR) and mountain (lower panel: Molina de Aragón, MOL).

## Model assessment

Modeled daily wind speed at 10-m is compared to the measurements of 28 ECA&D meteorological stations. WRF-ERA40 both over and underestimates surface wind speed, being largest the deviation in mountain regions or sea-land interface regions. The complex topography of Catalunya clearly influences the surface winds and the dominant direction is only captured in around 40% of the analyzed cases.

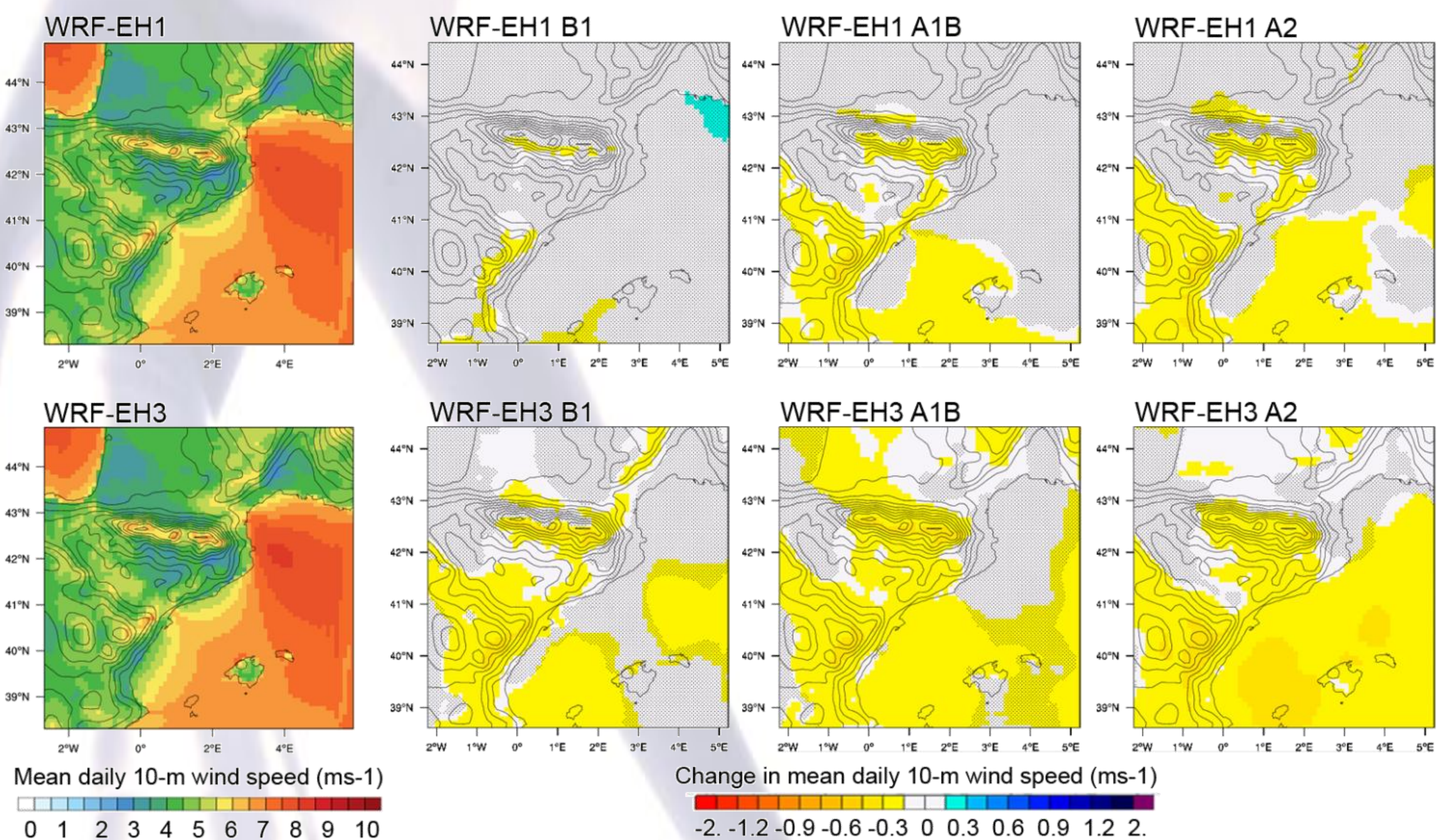
WRF-ERA40 reproduces adequately the geographical pattern of the wind speed at 60 m agl currently used for the definition of the wind farms preferential locations, although it yields higher wind intensities than the Departament of Territory and Sustainability (DTS) database.



Mean daily wind speed at 60 m agl ( $\text{ms}^{-1}$ ) for 1981-2000 as derived from WRF-ERA40 (left) –10-km horizontal resolution– and the DTS database (right) –200-m horizontal resolution –.

## Surface daily mean wind speed projections (2031-2050 vs 1981-2000)

There is a clear signal pointing towards a decrease in mean wind speed for the southern Mediterranean coast and mountain ranges (the Pyrenees and the Iberian System). Reductions up to 8.5% of mean wind intensity are detected, both overland and overseas, in the most intensive scenario (A2). The dominant wind direction at point locations remains unchanged, as well as the distribution of daily mean wind intensity.

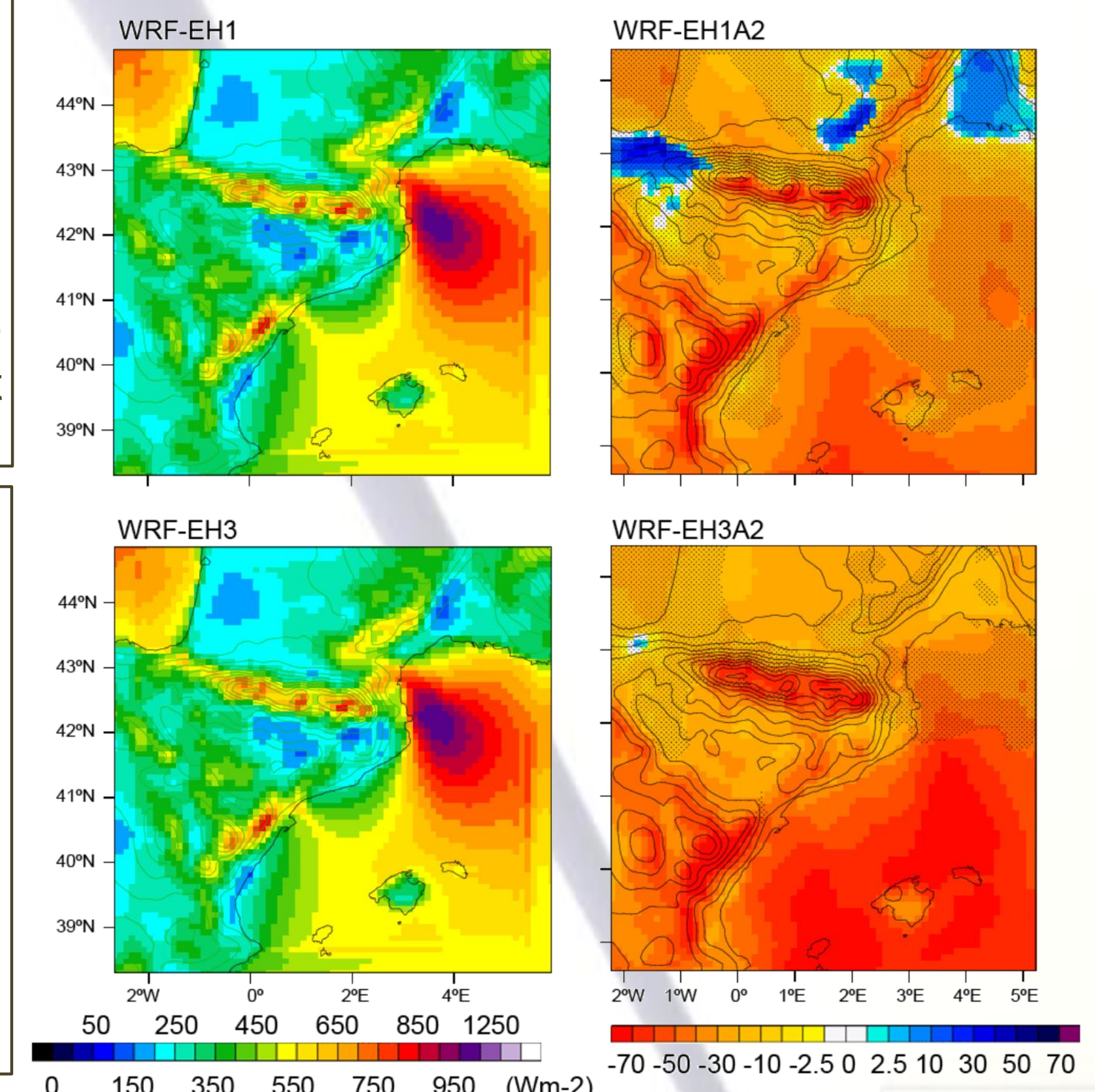


## Wind energy density projections (2031-2050 vs 1981-2000)

Wind energy density ( $E$ ,  $\text{W}\cdot\text{m}^{-2}$ ) is a function of the air density ( $\text{kg}\cdot\text{m}^{-3}$ ) and mean wind speed ( $\text{m}\cdot\text{s}^{-1}$ ).

$$E = \frac{1}{2} \rho v^3$$

WRF-EH1A2 and WRF-EH3A2 simulations suggest important reductions of wind energy density at 60 m agl for mid-21st century when compared to the control period (up to 20%). These changes could lead to significant variations in the zones characterized as preferential sources for the wind mills implementation at present (see figure on the right).



## Conclusions

- A future weakening of wind speed at surface is projected to happen, especially for mountain regions and southern Mediterranean coast. However, the dominant wind direction is expected to remain unchanged, as well as the distribution of daily mean wind intensity.
- Future changes in wind speed at hub-height could produce substantial changes in the wind energy density.
- Recommendation: Use dynamic procedures forced by climate change scenarios to define future preferential wind farms location.

## Acknowledgments:

We acknowledge the data providers in the ECA&D project. (Klein Tank et al., 2002). Data and metadata available at <http://www.ecad.eu>. The DTS from the Generalitat de Catalunya for providing the wind speed data at 60 m agl. The ECMWF for the ERA40 reanalyses; and the CERA (website of the World Data Center for Climate in Hamburg) for the ECHAM5/MPI-OM simulations. Background picture author P. Samo. We also want to thank the GECCC organization for the funding associated with this poster presentation.